# Image Quality and Robustness in Face Recognition: A Comparison of Classical and Deep Approaches

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**Abstract** :

This work investigates the impact of image quality on face recognition by comparing three representative approaches: Eigenfaces/PCA, Local Binary Pattern Histograms (LBPH), and a pretrained FaceNet model. The systems are evaluated under the same LFW verification protocol, first on clean images and then under progressively increasing Gaussian blur, Gaussian noise, brightness variation, and resolution reduction. Performance is measured through genuine and impostor comparisons using the Equal Error Rate (EER). A sharpness-based quality-aware rejection strategy is also evaluated to identify unreliable probe samples before matching. FaceNet achieves the best performance under all tested conditions, while the classical approaches exhibit different degradation-specific weaknesses. On mixed-quality probes, rejecting the lowest-quality 30% reduces the FaceNet EER from 4.85% to 1.84%.

## 1 Experimental Setup

The experimental framework was implemented in Python 3.10 to compare three complementary face-recognition approaches—Eigenfaces/PCA, Local Binary Pattern Histograms (LBPH), and a pre-trained FaceNet model—under the same acquisition conditions and verification protocol.

**Software.** The main tools used in the implementation are summarised in Table 1.

**Table 1**
Main software components and their role.

| Component | Tool / library |
|---|---|
| Image processing and degradations | OpenCV, NumPy |
| Dataset and PCA | scikit-learn |
| LBPH descriptor | NumPy, OpenCV |
| Deep face embeddings | `facenet-pytorch` |
| Data and experimental splits | pandas |
| Metrics and plots | NumPy, Matplotlib |

**Hardware.** All experiments were executed locally on CPU. The FaceNet network is used only as a fixed feature extractor, with no training or fine-tuning, making the complete pipeline practical without GPU acceleration.

**Reproducibility.** Dependencies are installed in an isolated virtual environment and recorded in a `requirements.txt` file. All random operations use a fixed seed of 42. The LFW training, gallery, and probe partition is generated once and stored in a CSV file, so every recognizer is evaluated on exactly the same biometric samples. Stochastic degradations, such as Gaussian noise, are generated deterministically as well.

**Implementation architecture.** The pipeline includes data splitting, preprocessing, feature extraction, matching, evaluation, robustness tests, and quality-aware rejection. External libraries provide standard components such as PCA, image processing, and the pretrained FaceNet model, while the experimental protocol, degradation pipeline, matching/evaluation logic, and quality-aware analysis were developed for this project.

## 2 Dataset and Protocol

### *2.1 LFW Dataset*

The experiments are conducted on **Labeled Faces in the Wild (LFW)** Huang et al., 2007b, a public face-recognition dataset containing images acquired under unconstrained conditions. LFW includes natural variations in pose, expression, illumination, and acquisition conditions, making it suitable for evaluating recognition robustness under additional controlled image degradations.

The dataset is loaded through the scikit-learn interface using the *funneled* version Huang et al., 2007a, which provides approximately aligned face crops. Only identities with at least ten available images are retained, resulting in 4,324 available images from 158 subjects, with an image size of approximately $125 \times 94$ pixels.

Since the number of available images is strongly unbalanced across subjects, exactly ten images per identity are selected to prevent frequently represented subjects from dominating the evaluation.

### *2.2 Balanced Experimental Split*

The selected images are divided into three non-overlapping roles: five training images, one gallery image, and four probe images per subject. The resulting protocol is summarised in Table 2.

**Table 2**
Balanced experimental protocol on LFW.

| Subset | Per subject | Subjects | Images |
|---|---|---|---|
| Training | 5 | 158 | 790 |
| Gallery | 1 | 158 | 158 |
| Probe | 4 | 158 | 632 |
| Total | 10 | 158 | 1,580 |

No image is shared between training, gallery, and probe sets, so the protocol is **sample-disjoint**. The same identities appear in all three partitions, however, and the experiment is therefore not subject-disjoint. The objective is to compare the recognition approaches under a fixed identity population while progressively degrading only the probe acquisitions.

All recognizers use exactly the same gallery and probe sets, ensuring that performance differences are measured under identical biometric trials.

### 2.3 Verification Protocol

For verification evaluation, every probe template is compared with every gallery template. Each entry of the resulting $632 \times 158$ distance matrix is interpreted as an independent verification comparison: comparisons between samples of the same identity are **genuine**, whereas comparisons involving different identities are **impostor**.

Since the gallery contains one reference template for each of the 158 identities, each probe produces one genuine and 157 impostor comparisons. The complete experiment therefore contains

$$N_{\text{genuine}} = 632, \qquad N_{\text{impostor}} = 632(158-1) = 99,224.$$

All three recognizers produce distance scores, with lower values indicating a stronger match. For a decision threshold $t$, a comparison is accepted when $d \le t$. Performance is measured through the False Acceptance Rate (FAR) and False Rejection Rate (FRR):

$$FAR(t) = \frac{FA(t)}{N_{\text{impostor}}}, \qquad FRR(t) = \frac{FR(t)}{N_{\text{genuine}}}.$$

The main figure of merit used throughout the experiments is the **Equal Error Rate (EER)**, corresponding to the operating point where

$$FAR(t_{\text{EER}}) \approx FRR(t_{\text{EER}}).$$

A lower EER indicates better separation between genuine and impostor comparisons.

### 2.4 Preprocessing

The classical and deep approaches require different input representations. Eigenfaces and LBPH operate on 8-bit grayscale images at the native LFW size of approximately $125 \times 94$ pixels. Eigenfaces subsequently vectorises the image for PCA projection, whereas LBPH operates directly on its local grayscale structure.

FaceNet instead preserves RGB information and resizes each face to $160 \times 160$ pixels before applying the standard input normalisation required by the pre-trained network.

No additional face detector is used. Since LFW *funneled* already provides approximately aligned face crops, degradations are applied directly to these samples. This intentionally isolates the **recognition stage**, avoiding face-detection failures as an additional source of error.

## 3 Face Recognition Approaches

Three face-recognition approaches are evaluated, representing different feature paradigms: a global statistical representation (Eigenfaces), a hand-crafted local texture descriptor (LBPH), and a deep learned embedding (FaceNet). Their main characteristics are summarised in Table 3.

**Table 3**
Recognition approaches evaluated in the project.

| Method | Representation | Dim. | Distance |
|---|---|---|---|
| Eigenfaces | PCA | 171 | Euclidean |
| LBPH | Local texture | 16,384 | $\chi^2$ |
| FaceNet | Deep embedding | 512 | Euclidean |

### 3.1 Eigenfaces

The first recognizer implements the classical **Eigenfaces** approach Turk et al., 1991. Principal Component Analysis (PCA) is fitted exclusively on the 790 training images and represents each face by its projection onto a low-dimensional appearance subspace:

$$\mathbf{y} = \Phi_k^T(\mathbf{x} - \boldsymbol{\mu}),$$

where $\boldsymbol{\mu}$ is the training-set mean and $\Phi_k$ contains the selected principal directions.

The mean face $\boldsymbol{\mu}$ is obtained by averaging all 790 training images pixel by pixel. PCA is applied to the centred images $\mathbf{x} - \boldsymbol{\mu}$, and the columns of $\Phi_k$ correspond to the leading principal components. When reshaped into image form, these components are the *eigenfaces*, which represent the dominant directions of appearance variation in the training set.

The number of components is chosen to preserve at least 95% of the training variance, resulting in a 171-dimensional template. Gallery and probe images are projected using the same PCA transformation, without using evaluation samples during fitting. Templates are compared through Euclidean distance.

Eigenfaces provides a global statistical baseline, but the dominant directions captured by PCA may reflect appearance variations such as illumination, expression, and pose in addition to identity.

### 3.2 Local Binary Pattern Histograms

The second recognizer uses **Local Binary Patterns (LBP)** Ojala et al., 2002; Ahonen et al., 2006, a hand-crafted descriptor that encodes local texture through relative intensity comparisons between neighbouring pixels.

To preserve spatial information, the LBP representation is divided into an $8 \times 8$ grid. A normalised 256-bin histogram is extracted from each region and the histograms are concatenated, producing a 16,384-dimensional template. Unlike Eigenfaces, LBPH requires no fitting stage and applies the same fixed transformation independently to every image.

Histogram templates are compared using the Chi-square distance. Because LBP depends mainly on local intensity ordering rather than absolute values, it can tolerate some illumination changes, while pixel-level perturbations may directly alter the encoded patterns.

### 3.3 FaceNet

The third recognizer is based on **FaceNet** Schroff et al., 2015. The implementation uses the `InceptionResnetV1` network provided by `facenet-pytorch`, with weights pre-trained on VGGFace2 Cao et al., 2018. No training or fine-tuning is performed on the LFW samples; the network is used as a fixed feature extractor.

Each face is mapped to a 512-dimensional embedding that is $L_2$-normalised before matching. FaceNet is trained to place images of the same identity close in the embedding space and different identities farther apart through a triplet-loss objective Schroff et al., 2015. Gallery and probe embeddings are compared using Euclidean distance.

Unlike Eigenfaces and LBPH, FaceNet therefore relies on a representation learned specifically for identity discrimination from a large external face dataset.

### *3.4 Comparison of the Representations*

The three approaches provide complementary representations: Eigenfaces models global appearance from the experimental training set, LBPH extracts predefined local texture statistics, and FaceNet uses a deep identity-oriented representation learned from large-scale external data. The objective is not to equalise their training conditions, but to compare representative classical, handcrafted, and deep approaches under the same gallery, probe degradations, and verification protocol.

## 4 Baseline Results

Before applying any image degradation, the three recognizers are evaluated on the original probe images to establish the clean reference performance for the subsequent robustness experiments.

Table 4 reports the mean genuine and impostor distances together with the Equal Error Rate (EER).

**Table 4**
Clean-image verification performance on the LFW protocol.

| Method | Mean genuine | Mean impostor | EER |
|---|---|---|---|
| Eigenfaces | 5470.97 | 6067.01 | 42.96% |
| LBPH | 25.53 | 27.49 | 39.20% |
| FaceNet | 0.699 | 1.375 | **1.58%** |

The two classical approaches provide limited discrimination under the adopted protocol. Eigenfaces reaches an EER of 42.96%, while LBPH performs moderately better at 39.20%. FaceNet shows a substantially different behaviour, reducing the EER to 1.58% and providing much stronger separation between genuine and impostor comparisons.

Figure 1 illustrates the strong overlap between genuine and impostor distances for Eigenfaces, which explains its high verification error.

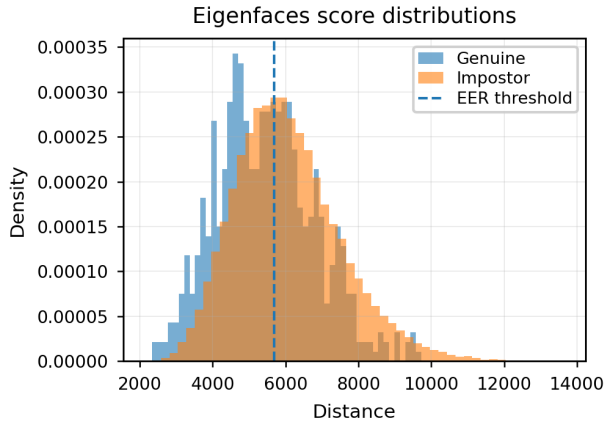



**Figure 1.** Genuine and impostor distance distributions for the Eigenfaces baseline.

The clean EER values are therefore used as the reference baseline for the subsequent degradation experiments.

## 5 Robustness Evaluation

The robustness experiments evaluate how the three face representations respond when probe quality is progressively degraded. The gallery remains clean in every experiment, while each degradation is applied to the original probe image before recognizer-specific preprocessing. Genuine and impostor scores are recomputed at every degradation level and summarised through the corresponding EER.

This protocol models a controlled-enrollment scenario in which the stored reference is acquired under favourable conditions while the subsequent recognition sample may be degraded. Applying each degradation before recognizer-specific preprocessing also ensures that all three methods receive representations derived from exactly the same degraded source image.

Four quality factors are considered: Gaussian blur, Gaussian noise, brightness variation, and resolution reduction. They affect complementary image properties: blur suppresses spatial detail, noise introduces random local intensity perturbations, brightness modifies the global intensity distribution, and resolution reduction removes spatial information through downsampling. Since each factor is varied independently, its effect on recognition can be analysed separately.

### *5.1 Gaussian Blur*

Gaussian blur is applied with $\sigma \in \{0, 0.5, 1, 2, 4\}$, where $\sigma = 0$ corresponds to the clean probe.

FaceNet remains close to its clean performance under mild and moderate blur: its EER increases from 1.58% to 2.23% at $\sigma = 2$. At $\sigma = 4$, however, it rises sharply to 11.03%, indicating that severe removal of fine facial structure affects the learned embedding.

LBPH varies more moderately, from 39.20% to 40.76%, whereas Eigenfaces remains close to its already high clean error. The three representations therefore react differently to the progressive loss of high-frequency facial information.

Figure 2 illustrates this behaviour across all tested blur levels, showing the progressive EER variation of the three recognizers as $\sigma$ increases.

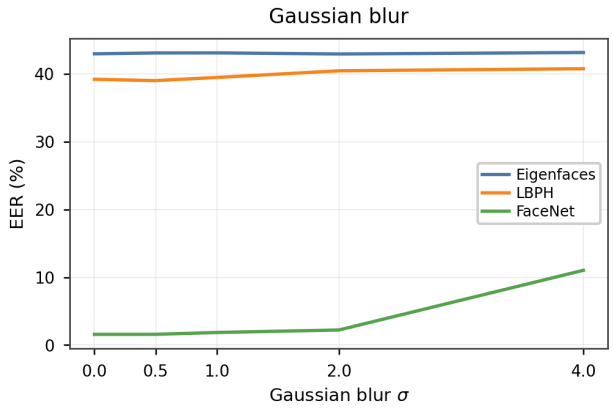



**Figure 2.** EER under increasing Gaussian blur.

### *5.2 Gaussian Noise*

Zero-mean additive Gaussian noise is applied to images in the $[0, 1]$ range using $\sigma \in \{0, 0.02, 0.05, 0.10, 0.20\}$, with pixel values clipped to the valid range.

Noise has the strongest effect on LBPH: its EER increases from 39.20% to 49.80% at $\sigma = 0.20$. This behaviour is consistent with the descriptor relying on local intensity comparisons, which can be directly altered by random pixel-level perturbations.

FaceNet remains accurate under moderate noise, reaching 3.96% EER at $\sigma = 0.10$, but severe noise increases the error to 20.83%. The small decrease from 1.58% to 1.42% at $\sigma = 0.02$ is considered experimental fluctuation rather than evidence of improved recognition.

Figure 3 shows the corresponding EER trends across the complete range of Gaussian noise levels.

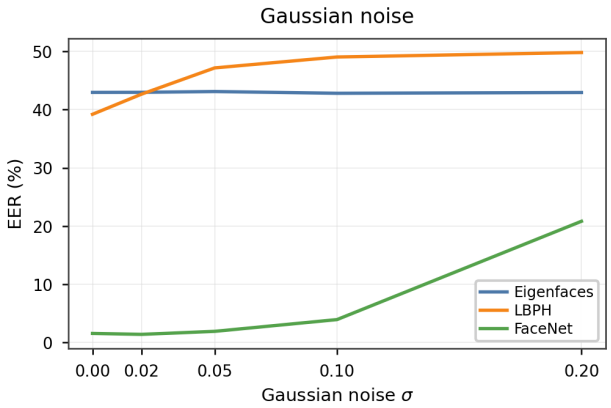



**Figure 3.** EER under increasing Gaussian noise.

### 5.3 Brightness Variation

Brightness is modified by multiplying image intensities by $\alpha \in \{0.50, 0.75, 1.00, 1.25, 1.50\}$ and clipping the result to the valid range. Values below one darken the image, while values above one increase brightness.

FaceNet is almost unaffected across the tested range, with EER values between 1.57% and 1.75%. LBPH is also comparatively stable, consistent with its dependence on relative local intensity ordering rather than absolute grayscale values.

Eigenfaces is substantially more sensitive to global intensity variation. Its EER rises from 42.96% to 48.23% at $\alpha = 0.50$ and to 46.42% at $\alpha = 1.50$. Darkening and brightening are not perfectly symmetric transformations, since increasing intensity can also introduce saturation through clipping.

Figure 4 summarises these effects across the tested brightness factors, highlighting the relative stability of FaceNet and LBPH and the greater variation observed for Eigenfaces.

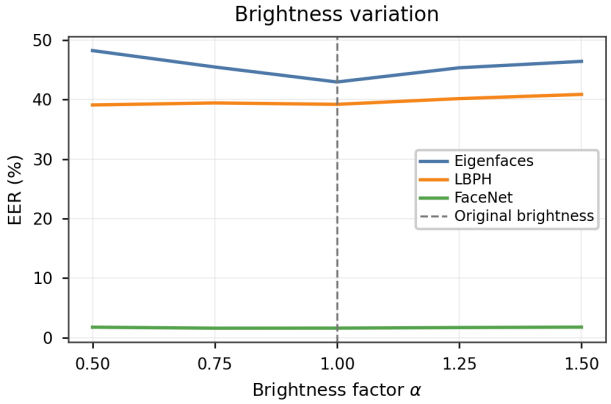



**Figure 4.** EER under brightness variation.

### 5.4 Resolution Reduction

Spatial resolution is reduced by downsampling each probe by a scale factor $s \in \{1.00, 0.75, 0.50, 0.25, 0.125\}$ and then resizing it back to its original dimensions. The recognizer therefore receives an image of the expected size, but information lost during downsampling cannot be recovered by interpolation.

FaceNet remains highly stable down to $s = 0.25$, corresponding to approximately $31 \times 23$ effective pixels, where its EER is still 2.57%. At $s = 0.125$, approximately $15 \times 11$ effective pixels, the EER rises sharply to 16.98%.

LBPH also deteriorates at the strongest reduction, reaching 41.85%. Eigenfaces remains close to its clean error level, although this occurs around an already weak baseline. Substantial spatial information can therefore be removed before the deep representation fails, whereas extreme downsampling eventually eliminates detail required for reliable matching.

Figure 5 reports the EER evolution over all tested resolution scales and makes the sharp degradation at the lowest resolution particularly visible.

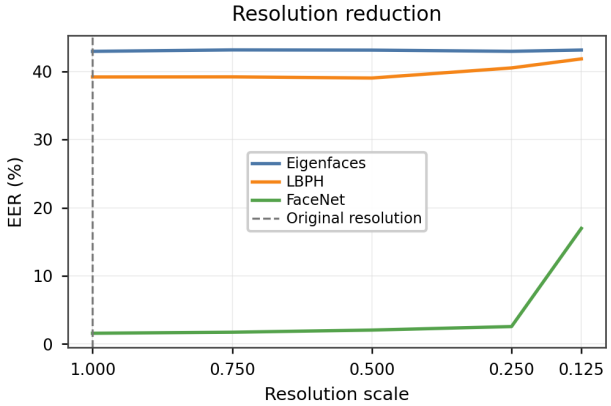



**Figure 5.** EER under progressive resolution reduction.

### 5.5 Overall Robustness Comparison

Taken together, Figures 2–5 highlight the degradation-specific robustness patterns of the three recognition approaches.

For direct numerical comparison, Table 5 reports the clean baseline together with one representative severe condition from each degradation experiment.

**Table 5**
EER (%) under selected severe degradation conditions.

| Condition | Eigen. | LBPH | FaceNet |
|---|---|---|---|
| Clean | 42.96 | 39.20 | **1.58** |
| Blur $\sigma = 4$ | 43.15 | 40.76 | **11.03** |
| Noise $\sigma = 0.20$ | 42.94 | 49.80 | **20.83** |
| Brightness $\alpha = 0.50$ | 48.23 | 39.09 | **1.74** |
| Resolution $s = 0.125$ | 43.15 | 41.85 | **16.98** |

The comparison reveals degradation-specific failure modes. LBPH is particularly vulnerable to Gaussian noise but remains comparatively stable under brightness variation. Eigenfaces is most clearly affected by global intensity changes. FaceNet achieves the lowest absolute EER in every tested condition and tolerates moderate degradation well, but severe blur, noise, and resolution loss still produce substantial performance degradation.

### *5.6 Sensitivity Relative to Clean Performance*

Absolute EER describes recognition reliability at a given quality level, but it does not directly quantify how strongly performance has changed relative to the clean condition. To complement the absolute comparison, degradation sensitivity is defined as

$$\Delta\text{EER} = \text{EER}_{\text{degraded}} - \text{EER}_{\text{clean}}. \tag{1}$$

Table 6 reports this variation for the same severe conditions considered above.

**Table 6**
EER variation in percentage points relative to clean performance.

| Condition | ΔEigen. | ΔLBPH | ΔFaceNet |
|---|---|---|---|
| Blur $\sigma = 4$ | +0.19 | +1.56 | +9.45 |
| Noise $\sigma = 0.20$ | -0.02 | +10.60 | +19.25 |
| Brightness $\alpha = 0.50$ | +5.27 | -0.11 | +0.16 |
| Resolution $s = 0.125$ | +0.19 | +2.65 | +15.40 |

The sensitivity view reinforces the degradation-specific behaviour. Noise produces the largest deterioration for LBPH, whereas global brightness variation is the dominant perturbation for Eigenfaces. FaceNet is nearly invariant to brightness but exhibits substantial performance loss under severe noise, resolution reduction, and blur.

These variations must nevertheless be interpreted together with the absolute EER values in Table 5. FaceNet can show a larger ΔEER because it starts from a very low clean error, yet it remains substantially more accurate than the classical approaches under every tested condition. Conversely, the small variations exhibited by Eigenfaces under blur, noise, and resolution do not indicate superior robustness, since its baseline discrimination is already weak.

Absolute EER and ΔEER therefore provide complementary information: the former describes the reliability of the biometric system at a particular quality level, while the latter measures its sensitivity to changing acquisition conditions. Considering both prevents a weak but nearly flat recognizer from being mistaken for a genuinely robust system.

## 6 Quality-Aware Recognition

The robustness experiments show that severe degradation can reduce recognition performance even for the deep representation. This motivates a quality-aware strategy in which unreliable probe samples are identified before matching rather than forcing a recognition decision for every acquisition. Quality information can be used to identify biometric samples that should not be submitted directly to recognition Marsico et al., 2011.

The proposed experiment uses image sharpness as a simple quality indicator. The objective is not to construct a general Face Image Quality Assessment model, but to evaluate whether a degradation-related quality measure can improve recognition reliability under mixed-quality acquisition conditions.

### *6.1 Sharpness-Based Quality Measure*

Gaussian blur suppresses high-frequency information and reduces local intensity variations. Following a sharpness quality formulation based on neighbouring-pixel intensity differences Abboud et al., 2012, probe quality is estimated from the average absolute difference between neighbouring grayscale pixels.

For an image $I$ represented in the range $[0, 1]$,

$$Q_{\text{sharp}} = \frac{1}{2}\big(\,\text{mean}_{x,y}\,|I(x+1,y) - I(x,y)| + \text{mean}_{x,y}\,|I(x,y+1) - I(x,y)|\big). \tag{2}$$

Higher values indicate stronger local intensity variation and, generally, a sharper image. The score is computed directly from each probe before recognition and is independent of the selected recognizer.

To verify that the measure responds consistently to blur, the 632 original probes are evaluated at $\sigma \in \{0, 0.5, 1, 2, 4\}$. Figure 6 shows a clear monotonic decrease in mean sharpness as blur increases.

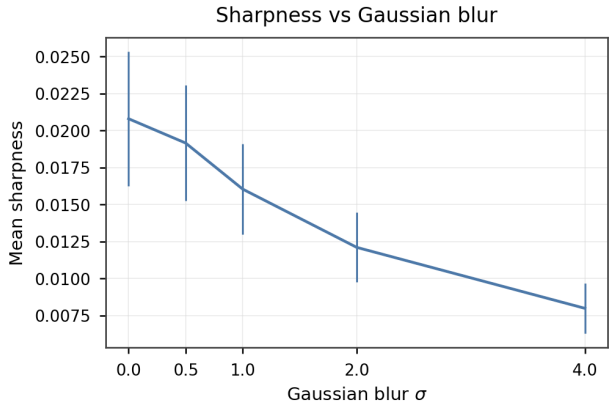



**Figure 6.** Mean sharpness score under increasing Gaussian blur. Error bars represent one standard deviation across probe samples.

Quantitatively, the mean quality score decreases from 0.020790 for clean probes to 0.007985 at $\sigma = 4$, corresponding to a reduction of approximately 61.6%. This provides a substantial dynamic range for ranking samples according to blur-related quality, despite the sample-dependent variability visible in the error bars.

The score distributions are not perfectly separated because the original face images naturally contain different amounts of texture and detail. The quality mechanism therefore operates on measurements obtained from each image rather than directly using the known synthetic blur level.

### *6.2 Mixed-Quality Rejection Protocol*

The five blur conditions form a mixed-quality set of 3,160 probes. Samples are ranked from lowest to highest $Q_{\text{sharp}}$, and progressively larger fractions $r \in \{0, 10, 20, 30, 40, 50\}\%$ are rejected.

The known blur level is never used to make the rejection decision; it is retained only for subsequent diagnostic analysis. At each rejection rate, all genuine and impostor comparisons associated with discarded probes are removed and the EER is recomputed on the retained samples.

The experiment therefore measures an **error–rejection trade-off**: increasing the quality threshold may improve recognition reliability, but reduces the fraction of acquisitions accepted for biometric processing.

### *6.3 Quality-Aware Results*

Figure 7 shows how the EER changes as progressively larger fractions of low-sharpness probes are removed, while Table 7 reports the corresponding numerical values and the number of retained acquisitions.

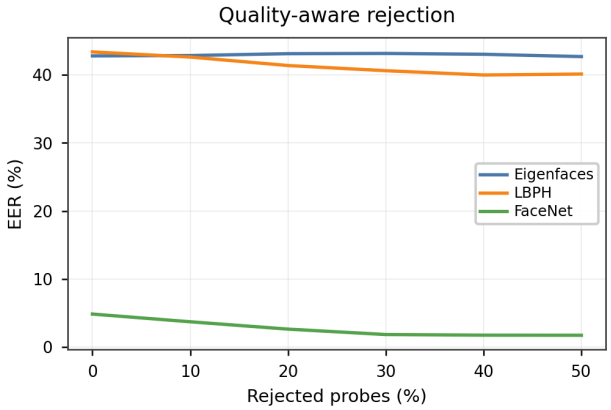



**Figure 7.** EER as progressively larger fractions of the lowest-sharpness probes are rejected.

**Table 7**
Quality-aware rejection results on the mixed-blur probe set. EER values are reported in percent.

| Rejected | Retained | Eigen. | LBPH | FaceNet |
|---|---|---|---|---|
| 0% | 3160 | 42.80 | 43.39 | 4.85 |
| 10% | 2844 | 42.86 | 42.63 | 3.72 |
| 20% | 2528 | 43.13 | 41.39 | 2.64 |
| 30% | 2212 | 43.16 | 40.62 | **1.84** |
| 40% | 1896 | 43.04 | 39.99 | 1.74 |
| 50% | 1580 | 42.70 | 40.13 | 1.73 |

FaceNet shows the strongest benefit. Its EER decreases from 4.85% when the complete mixed-quality set is accepted to 1.84% at 30% rejection, already close to the 1.58% clean baseline. The first 30% of rejected samples therefore contain a large fraction of the degradation responsible for the increased verification error.

Beyond this point, the curve exhibits clear diminishing returns: increasing rejection to 40% and 50% lowers the FaceNet EER only to 1.74% and 1.73%, respectively. The 30% operating point should therefore be interpreted as an empirical compromise for this mixed-blur experiment rather than as a universal quality threshold.

The availability cost is also substantial. Increasing rejection from 30% to 50% removes 632 additional probes, reducing the retained set from 2,212 to 1,580 samples, while improving the FaceNet EER by only 0.11 percentage points. The additional loss of available acquisitions is therefore large compared with the marginal recognition benefit.

LBPH also benefits from quality filtering, decreasing from 43.39% to approximately 40%. Eigenfaces instead remains close to 43% throughout the experiment, indicating that removing blurred samples cannot compensate for its weak baseline discrimination.

### *6.4 Analysis of Rejected Samples*

Improved EER alone does not demonstrate that the quality measure is actually identifying the intended low-quality acquisitions. To assess the selectivity of the rejection mechanism, Figure 8 shows how the rejected samples are distributed across blur levels as the rejection rate increases, while Table 8 provides a detailed snapshot at the 30% operating point.

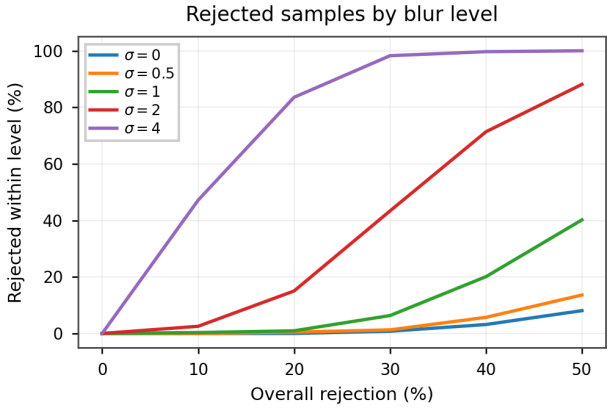



**Figure 8.** Fraction rejected within each Gaussian-blur group as the overall quality rejection rate increases.

**Table 8**
Composition of the rejected set at 30% overall rejection. Each blur group contains 632 probes.

| Blur $\sigma$ | Rejected | Within level |
|---|---|---|
| 0.0 | 5 | 0.79% |
| 0.5 | 8 | 1.27% |
| 1.0 | 40 | 6.33% |
| 2.0 | 274 | 43.35% |
| 4.0 | 621 | **98.26%** |
| Total | 948 | – |

At 30% overall rejection, almost all probes affected by the strongest blur are discarded: 621 of 632 samples at $\sigma = 4$ are removed, together with 43.35% of the $\sigma = 2$ group. In contrast, only five clean probes are rejected. The quality score therefore strongly concentrates the rejection on samples affected by severe blur rather than removing probes indiscriminately.

This selectivity gradually decreases as the rejection threshold becomes more aggressive. At 50% overall rejection, all $\sigma = 4$ probes are discarded, but 8.07% of the clean samples and 13.61% of the mildly blurred ($\sigma = 0.5$) probes are also removed. Once most severely degraded acquisitions have been filtered out, increasing the threshold therefore begins to sacrifice progressively more acceptable samples.

Taken together, the EER and rejection-composition analyses illustrate the central quality-aware trade-off: rejecting unreliable acquisitions can substantially improve recognition reliability, but excessive rejection reduces sample availability while providing only limited additional performance gains.

## 7 Discussion and Limitations

The experiments show that recognition robustness must be interpreted together with absolute recognition performance and with the specific degradation affecting the biometric sample. The three feature paradigms exhibit different failure modes, while FaceNet maintains the lowest absolute EER under every tested condition.

The comparison should not be interpreted as a controlled training-budget benchmark. Eigenfaces learns its PCA representation from the experimental training subset, LBPH is entirely hand-crafted, and FaceNet benefits from large-scale external pretraining. The aim is instead to compare representative classical, local hand-crafted, and deep representations under identical probe degradations and verification trials.

The quality-aware experiment also has a deliberately limited scope. Sharpness is well suited to detecting blur because blurred images lose local intensity variation, but face-image quality is inherently multidimensional. Noise, illumination, pose, occlusion, compression, and face resolution cannot be reliably described by sharpness alone; indeed, noise may even increase local intensity differences. The proposed rejection mechanism should therefore be considered a proof-of-concept quality gate rather than a general Face Image Quality Assessment model.

Several properties of the experimental protocol also limit the generality of the reported results. The split is sample-disjoint but not subject-disjoint, so the Eigenfaces PCA space is estimated from different images of identities that later appear during evaluation. The gallery contains only one reference template per identity, whereas multiple enrollment samples or template aggregation could improve recognition stability.

Moreover, the tested degradations are synthetic and applied independently. Real acquisitions may combine motion blur, illumination variation, compression, pose changes, and sensor noise. Finally, the use of pre-cropped and approximately aligned LFW *funneled* images isolates the recognition stage, so the reported results do not include degradation-induced failures of face detection or alignment.

A more complete evaluation could therefore adopt a subject-disjoint protocol, multiple enrollment templates, combined or naturally occurring degradations, and a multidimensional or learned face-quality estimator.

## 8 Conclusion

This project evaluated the robustness of three representative face-recognition approaches—Eigenfaces, Local Binary Pattern Histograms (LBPH), and FaceNet—under controlled image-quality degradations on LFW.

The experiments show that robustness is strongly representation- and degradation-dependent. FaceNet achieves the best absolute performance in every tested condition, with a clean EER of 1.58%, while severe blur, Gaussian noise, and resolution loss still produce substantial degradation. The classical approaches exhibit different failure modes, with LBPH particularly sensitive to noise and Eigenfaces more affected by global brightness variation.

The quality-aware experiment further shows that sample quality can be used operationally to improve recognition reliability. On the mixed-blur probe set, rejecting the lowest-quality 30% according to a sharpness measure reduces the FaceNet EER from 4.85% to 1.84%, while discarding fewer than 1% of the clean probes.

Overall, the results indicate that robust biometric recognition requires both a discriminative representation and an explicit assessment of whether the acquired sample contains sufficient reliable information for matching.

## References

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